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The Many Interpretations of Quantum Mechanics

By Graham P. Collins on November 19, 2007

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What is the ultimate nature of reality?

Are quantum effects constantly carving us into innumerable copies, each copy inhabiting a different version of the universe? Or do all those other worlds pop out of existence as mere might-have-beens? Do our particles surf on quantum waves? Or are we ultimately made of the quantum waves alone? Or do the waves merely represent how much information we could possess about the state of the world? And if the waves are just a kind of information, information about *what*? Or is the information all that there is—and all that we are?

Those are the kind of questions in play when a physicist tackles the dry-sounding issue of, “what is the correct interpretation of quantum mechanics?” About 80 years after the original flowering of quantum theory, physicists still don’t agree on an answer.

And although quantum mechanics is primarily the physics of the very small—of atoms, electrons, photons and other such particles—the world is made up of those particles. If their individual reality is radically different from what we imagine then surely so too is the reality of the pebbles, people and planets that they make up.

As recounted by our December article, [The Many Worlds of Hugh Everett](#) by journalist [Peter Byrne](#), 50 years ago the iconoclastic physics student Hugh Everett introduced the idea that quantum physics is incessantly splitting the universe into alternate branches. Byrne’s article talks about Everett’s life (did you know his son is the lead singer of the rock band *Eels*?) as well as about his theory and the “Copenhagen Interpretation” he aimed to supplant. But many other interpretations of quantum mechanics exist, and today Copenhagenists have more subtle variants to choose from than the one that Everett once called “a philosophic monstrosity.” Here is an all-too-short run-down on some of them.

The basic scenario an interpretation must address is when a quantum system is prepared in a combination of states known as a superposition. For example, a particle can be at both location A and B, or in the infamous thought experiment, Schrödinger's quantum cat can be alive and dead at the same time. The problem is that when we observe or measure a superposition, we get but one result: our detector reports either "A" or "B," not both; the cat would appear either very alive or very dead.

Copenhagen Interpretation

This interpretation (or variants of it) has long been the party line for quantum physicists. The Schrödinger equation describes how a wave function evolves smoothly and continuously over time, up until the point when our big, clunky measuring apparatus intervenes. The wave function enables us to predict, say, there's a 60% probability we'll detect the particle at location A. After we detect it at A or B, we have to represent the particle with a new wave function that conforms with the measurement result.

What bothers some people about this interpretation is the random, abrupt change in the wave function, which violates the Schrödinger equation, the very heart of quantum mechanics. Everett argued that this approach was philosophically a mess: it used two contradictory conceptual schemes to describe reality, the quantum one of wave functions and the classical one of us and our apparatus.

Many Worlds Interpretation

Everett's theory. Also known as the relative state formulation.

The superposition of the particle spreads to the apparatus, and to us looking at the apparatus, and ultimately to the entire universe. The components of the resulting superposition are like parallel universes: in one we see outcome A, in another we see outcome B. All the branches coexist simultaneously, but because they are completely non-interacting the "A" copy of us is completely unaware of the "B" copy and vice versa. Mathematically, this universal superposition is what the Schrödinger equation predicts if you describe the whole universe with a wave function.

What bothers people about this interpretation is its conclusion that we are perpetually dividing into multiple copies, which may have ghastly implications as well as being bizarre.

Bohmian Interpretation

Also known as the De Broglie–Bohm interpretation or the pilot wave interpretation.

This theory postulates that every particle not only has a wave function but also exists as an actual particle riding along at some precise but unknown location on the wave and being guided by it. How the wave guides the particle is described by a new equation that is introduced to accompany the standard Schrödinger equation. The randomness of quantum measurements comes about because we cannot know exactly where a particle started out. The

theory was proposed by David Bohm in 1952 (a few years before Everett's theory), extending a theory of Louis De Broglie's from 1927.

Changing the Rules

Some theorists seek to find a mechanism that causes the “collapse” of the wave function from a superposition of possibilities to a single outcome. For example, Roger Penrose has proposed that gravitational effects may play this role. Other models, such as the Ghirardi-Rimini-Weber theory, introduce specific modifications to the Schrödinger equation. By differing from standard quantum theory, such models in principle might be falsifiable by experiment (or conversely, standard theory could be falsified in their favor).

Decoherence Theory

This is not an interpretation, but it is an important element of the modern understanding of quantum mechanics. It expands upon the kind of mathematical analysis that led Everett to his interpretation, because it analyzes the effect that stray quantum interactions with the surrounding environment have on a system in a superposition. The chief conclusion is that the almost unstoppable loss of information through these channels “decoheres” a quantum superposition, making it more like an ordinary classical state. It explains very well why we see the classical world that we do, and clarifies the requirements to keep quantum effects manifest in the lab.

Copenhagenists can point to decoherence as an explanation of what makes large classical systems different from small quantum systems (in general, large systems decohere much more readily and rapidly than tiny ones). Everettians can point to it as a more complete explanation of how the parallel branches form and become independent. But best of all, decoherence can be studied experimentally, and a very active area of quantum research is confirming it and exploring it in ever greater detail.

Consistent Histories

This scheme analyzes sequences of states of a system (which may include the whole universe), to find what questions can be consistently answered about the system, such as “was the particle at A or B at time T?” The measurement problem, however, is not resolved: the question of which histories actually happen remains a matter of probabilities just as with the standard Copenhagenist approach.

Is it Real?

In some respects the decision between a Copenhagenist and an Everettian viewpoint boils down to a basic question: Is the wave function real or is it just information? If it is “real”—in some sense the universe really consists of quantum waves propagating around—then one tends to be driven to an Everettian viewpoint; the “collapses” that wave functions must undergo to produce the one reality that we see are too problematic. But if the wave function is just information, for example, a representation of what an experimenter knows about a system, then that “collapse” is completely natural. Imagine the standard classical scenario of

flipping a coin. Before you look at it, your knowledge of its state is “50% chance of heads, 50% chance of tails.” When you look, your knowledge instantaneously changes to, say, “100% heads, 0% tails.”

“Shut Up and Calculate!”

Some physicists talk of the “shut up and calculate interpretation”: ignore the philosophical puzzle of how the classical and the quantum coexist and use the Schrödinger equation (and all the subsequent mathematical developments of quantum theory) to compute quantities of practical interest. These include energy levels of atoms; predictions for particle collider experiments; the properties of semiconductors, superconductors and other materials; and so on. It is all that most physicists ever need.

Transactional Interpretation

This interpretation has waves traveling forward and backward in time, setting up standing waves, for example between an emitter of a particle and its subsequent detector. It was proposed by John G. Cramer (physicist and science fiction author) in 1986 and claimed by him to provide insight into puzzles such as wave function collapse and the Schrödinger’s cat experiment. These insights have led Cramer to pursue an experiment to try to demonstrate the sending of signals backward in time (which most quantum physicists will tell you is impossible if standard quantum mechanics is correct).

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